```
import numpy as np

import matplotlib.pyplot as plt

import pandas as pd

import seaborn as sns

%matplotlib inline

dataset = pd.read_csv(r"C:\Users\srini\Downloads\11th - mlr\11th -
mlr\MLR\House_data.csv")

dataset.head()

print(dataset.isnull().any())

print(dataset.dtypes)

dataset = dataset.drop(['id','date'], axis = 1)

with sns.plotting_context("notebook",font_scale=2.5):
    g =
sns.pairplot(dataset[['sqft_lot','sqft_above','price','sqft_living','bedrooms']],
                 hue='bedrooms', palette='tab20',height=6)
g.set(xticklabels=[]);


X = dataset.iloc[:,1:].values
y = dataset.iloc[:,0].values


from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)

y_pred = regressor.predict(X_test)

import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3, 4,5,6,7,8,9,10,11,12,13,14,15,16,17]]

regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()

import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3, 4,6,7,8,9,10,11,12,13,14,15,16,17]]

regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()
```

```
import statsmodels.api as sm
X_opt = X[:, [0, 1, 2, 3,6,7,8,9,10,11,12,13,14,15,16,17]]
regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
regressor_OLS.summary()
```